1. Bézout's identity:

$$\gcd(a, b) = d \implies \exists x, y \in \mathbb{Z} : ax + by = d. \tag{1}$$

   From (1) and $\gcd(a, b) = 1$ we have:

$$ax + by = 1$$

   By multiplying right part of $a|c$ by the equation above we get:

$$a \mid c \iff a \mid c(ax + by) \iff a \mid acx + bcy$$

   $a \mid acx$ is obviously true.
   From the statement: $a \mid bc \implies a \mid bcx$
   Therefore $a \mid c$

2. Let's assume:

$$S = \{p : \text{prime},\ p \equiv 3 \pmod 4\} = \{p_1, \ldots, p_N\}$$

   for some finite N.
   Let's consider the following product:

$$P = \prod_{i=1}^{N} p_i$$

   Let's consider two cases:

   (a)
$$P \equiv 1 \pmod 4$$

   In this case we can take $P' = P + 2$.

$$\forall i : 2 \le i \le N : \quad P' \equiv 2 \pmod{p_i}$$
$$P' \equiv 3 \pmod 4$$

(b)

$$P \equiv 3 \mod 4$$

In this case let's consider $P' = P * 3 + 2$. P' satisfies same criteria as in previous case.

Now let's consider arbitrary number $n = q_1 \dots q_m,$ where q_i are primes and it's prime factorization. If $n \equiv 3 \pmod 4$, then for its factors the following holds:

(a)

$$\forall x,\ x \mid n : x \neq 2 \pmod 4$$

(b)

$$\exists x, x \mid n : x \equiv 3 \pmod 4$$

Since P' is not divisible by any elements of S but it must have a factor that satisfies conditions above it means contradiction, therefore S is not finite.

3.